Vector Spaces

Recall from last class:

- **Theorem 3.3:** If A is an $n \times n$ matrix and E is an elementary matrix, then

  $$\det(EA) = \det(E)\det(A),$$

  and

  $$\det(E) = \begin{cases} 1 & \text{if } E \text{ represents adding a multiple of a row to another row,} \\ -1 & \text{if } E \text{ represents interchanging two rows,} \\ r & \text{if } E \text{ represents scaling a row by a factor of } r. \end{cases}$$

- **Theorem 3.6:** If A and B are n by n matrices, then

  $$\det(AB) = \det(A)\det(B).$$

- **Corollary:** A is invertible if and only if $\det(A) \neq 0$.

- **Theorem:** If A is a 2×2 matrix, then the area of the parallelogram determined by the columns of A is $|\det A|$. If A is a 3×3 matrix, then the volume of the parallelepiped is $|\det A|$.

- **Theorem:** Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be linear with associated matrix A. If S is a parallelogram in $\mathbb{R}^2$, then $\text{Area}(T(S)) = |\det A| \text{Area}(S)$.

Definition 1. *A real* **vector space** *is a non-empty set V on which two operations, scalar multiplication and addition, are defined subject to the following 10 axioms. We call elements in V vectors. Let $u, v, w \in V$ and $c, d \in \mathbb{R}$.*

1. $u + v \in V$.
2. $u + v = v + u$.
3. $(u + v) + w = u + (v + w)$.
4. *there exists a vector* 0_V *in* V *such that* $u + 0_V = u$.
5. *there exists a vector* $(-u)$ *in* V *such that* $u + (-u) = 0_V$.
6. $cu \in V$.
7. $c(u + v) = cu + cv$.
8. $(c + d)u = cu + du$.
9. $c(du) = (cd)u$.
10. $1 \cdot u = u$.

1. Determine if the following are vector spaces.

(a) $\mathbb{R}^n$ with the usual operations.

(b) $S = \{(y_0, y_1, \dots) : y_k \in \mathbb{R} \text{ for all } k \geq 0\}$. Define addition component-wise and scalar multiplication by scaling every component.

(c) Let $V = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 : x^2 + y^2 = 1 \right\}$ with the usual operations.

(d) Let $\mathbb{P}_n = \{a_0 + a_1 t + \cdots + a_n t^n : a_0, \ldots, a_n \in \mathbb{R}\}$ with operations defined by corresponding operations on polynomials.

(e) Let $M_{n\times n}$ be the set of $n \times n$ matrices with the operations coming from matrix addition and scaling a matrix.

(f) Let $W = \{(x, y, 1) : x, y \in \mathbb{R}\}$ with the usual operations.

(g) Let $Z = \{(a, b) : a, b \in \mathbb{R}\}$ and $(a, b) + (c, d) = (a + c, 0)$ and $r(a, b) = (ra, 0)$.

(h) Let $Y = \{(a, b) : a, b \in \mathbb{R}\}$ and $(a, b) + (c, d) = (a + c, b - d)$ and $r(a, b) = (ra, rb)$.

(i) Let $F = \{f : \mathbb{R} \to \mathbb{R} : f \text{ is a function}\}$ with operations coming from the usual operations on functions.

Definition 2. *A* **subsapce** *of a vector space V is a subset of V such that*

(a) $0_V \in H$.

(b) *H is closed under vector addition; i.e. if $u, v \in H$, then $u + v \in H$.*

(c) *H is closed under scalar multiplication; i.e. if $c \in \mathbb{R}$ and $u \in H$, then $cu \in H$.*

2. Let V be a vector space.

(a) Is $H = \{0_V\}$ a subspace of V?

(b) Suppose $V = \mathbb{R}^3$, then is $\mathbb{R}^2$ a subspace of V?

Theorem 4.1 If $v_1, \dots, v_p \in V$, then $\text{Span}\{v_1, \dots, v_p\}$ is a subspace of V. We call it the subspace generated by or spanned by $v_1, \dots, v_p$.

3. Prove the Theorem.

4. Consider the vector space of polynomials of degree 2 or less. We denote this vector space by $\mathbb{P}_2$.

(a) Is the set of polynomials of degree 2 a subspace?

(b) Let $u = 1 + t$, $v = t$, and $w = 1 + t + 2t^2$. Is $w \in \text{Span}\{u, v\}$?

5. Let W_1 and W_2 be subspaces of a vector space V.

(a) When is $W_1 \cap W_2$ a subspace of V?

(b) When is $W_1 \cup W_2$ a subspace of V?